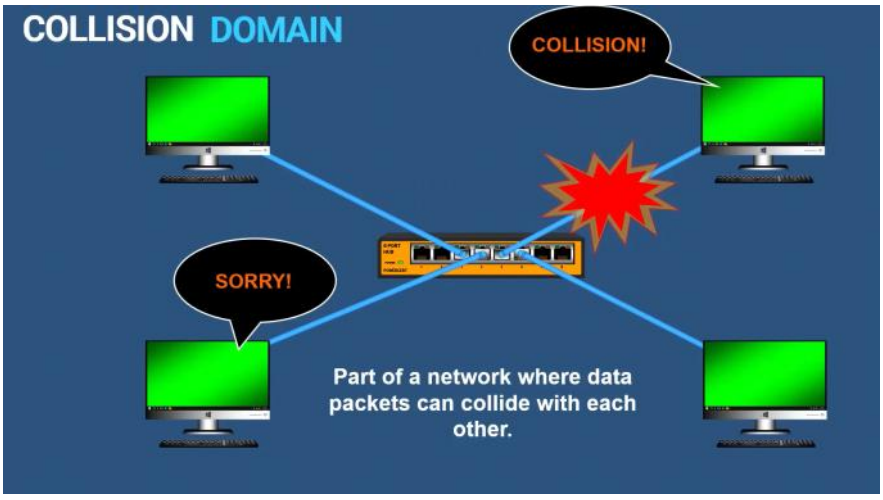
Collision Domains vs. Broadcast Domains

Friday, April 24, 2026 7:21 AM

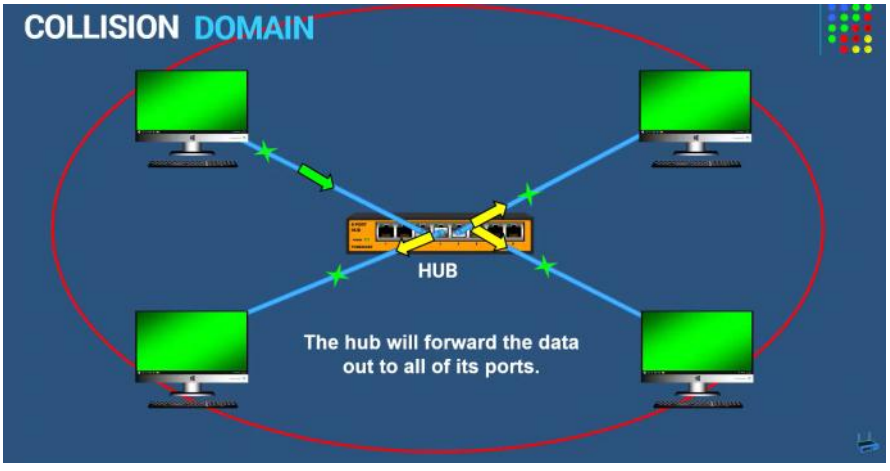
Collision Domains vs. Broadcast Domains

Collision Domains

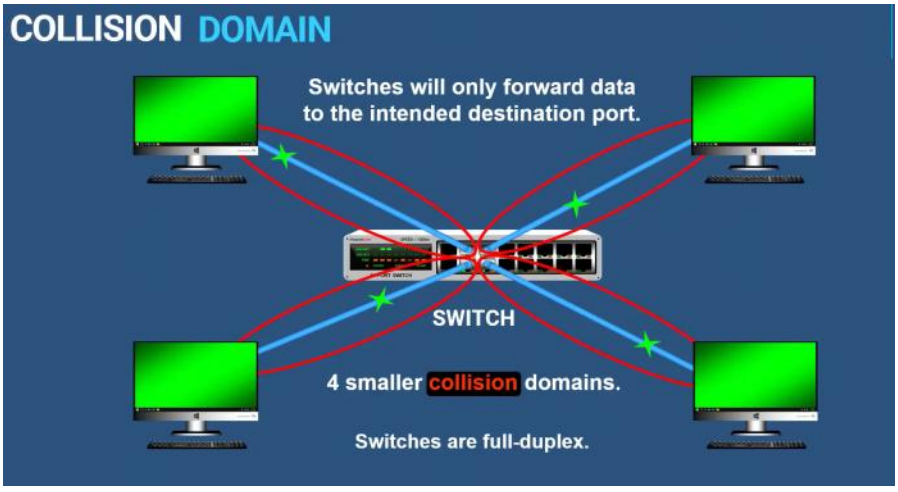
A collision domain is a network segment where data packets can collide if devices send data simultaneously. Collisions slow down network performance as devices must wait a random interval before resending data.



- Hubs: Operate in a single, large collision domain across all ports. They are half-duplex, meaning they can only send or receive data in one direction at a time, and they forward all incoming data to every connected port.

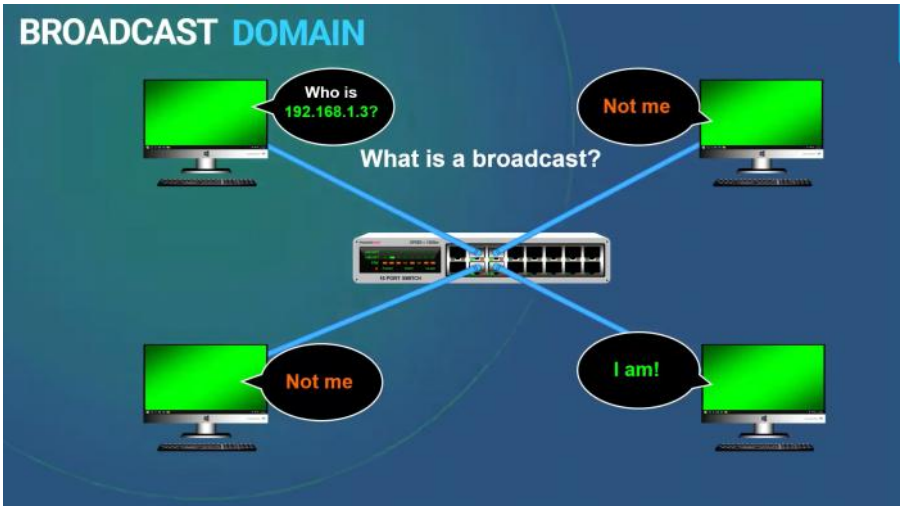


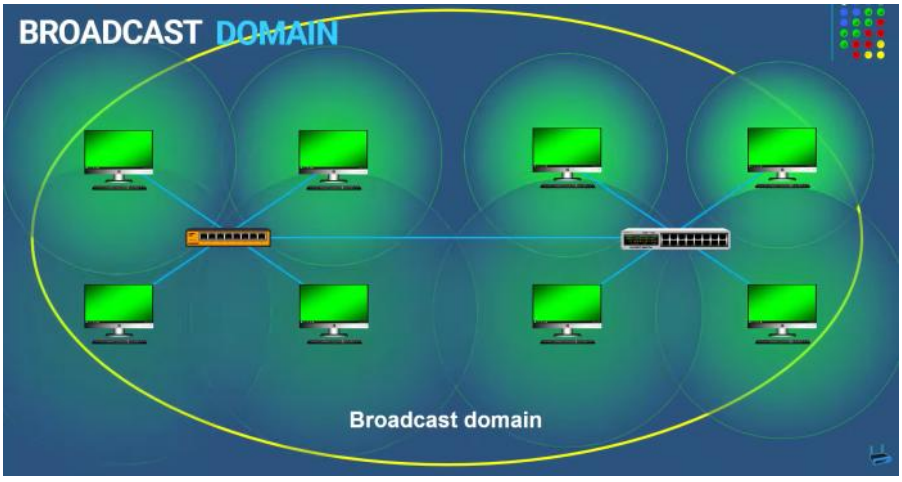
- Switches: Improve performance by creating smaller, individual collision domains for each port. They are full-duplex, allowing simultaneous communication in both directions, and forward data only to the intended destination port, significantly reducing or eliminating collisions.

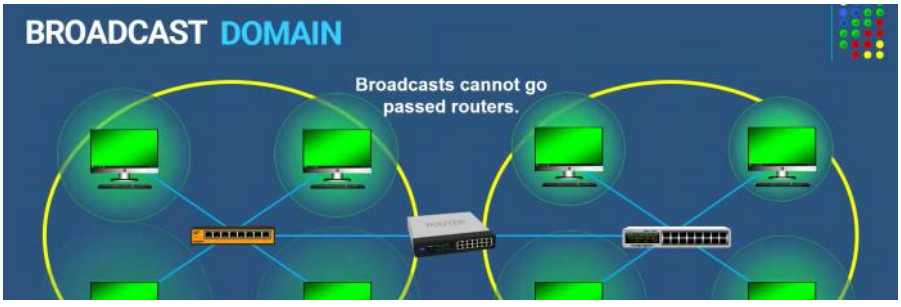


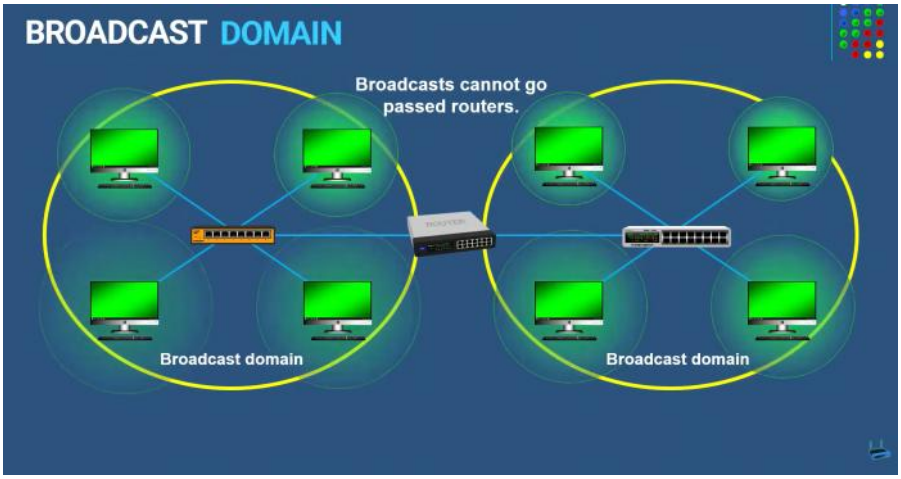
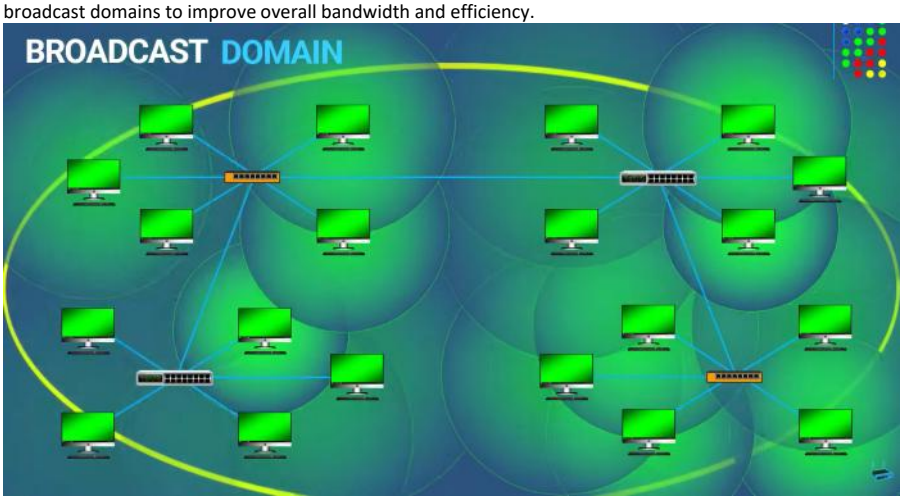
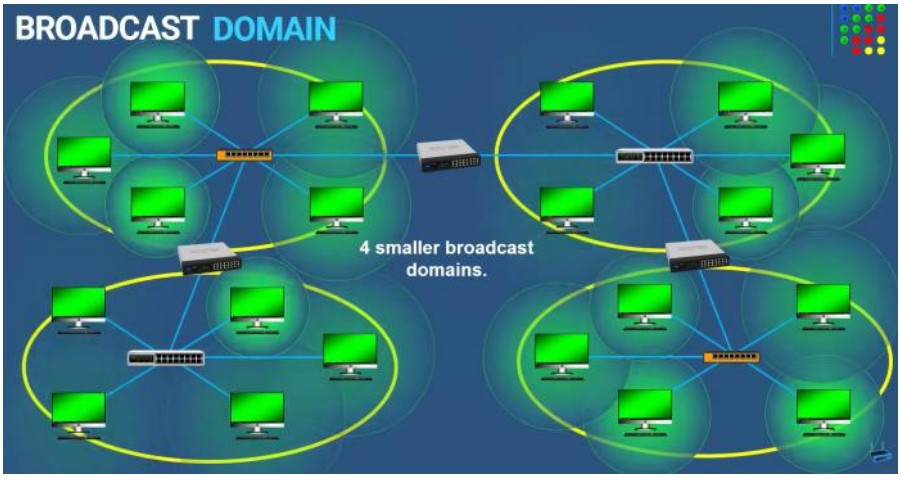
Broadcast Domains

A broadcast domain is a network area where devices receive broadcast traffic sent by other devices to locate them.









- Behavior of Hardware: Both hubs and switches forward broadcast traffic, meaning they do not divide or restrict broadcast domains.
- Routers: These devices block broadcast traffic. By placing routers between network segments, administrators can divide a large, congested network into multiple smaller broadcast domains to improve overall bandwidth and efficiency.